

Practice 3D Deconvolution Microscopy

<http://go.epfl.ch/ambia2026/>

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<http://bigwww.epfl.ch/deconvolution/deconvolutionlab2/>

DeconvolutionLab2

Installation

DeconvolutionLab2 is a versatile Java software package compatible with both ImageJ and Fiji.

DeconvolutionLab2 can function as a standalone application, seamlessly running on all OS such as Windows, Linux, and MacOSX.

<https://imagej.net/ij/>



- Download ImageJ IJ1
- Move ImageJ in your working place

OR

<https://fiji.sc/>



- Download Fiji
- Move ImageJ in your working place

<http://bigwww.epfl.ch/deconvolution/>

- Get DeconvolutionLab2.jar
- Get PSFGenerator.jar
- Download FFTW2.zip (optional)
- Copy theses files in the plugins folder



DeconvolutionLab2

DeconvolutionLab2

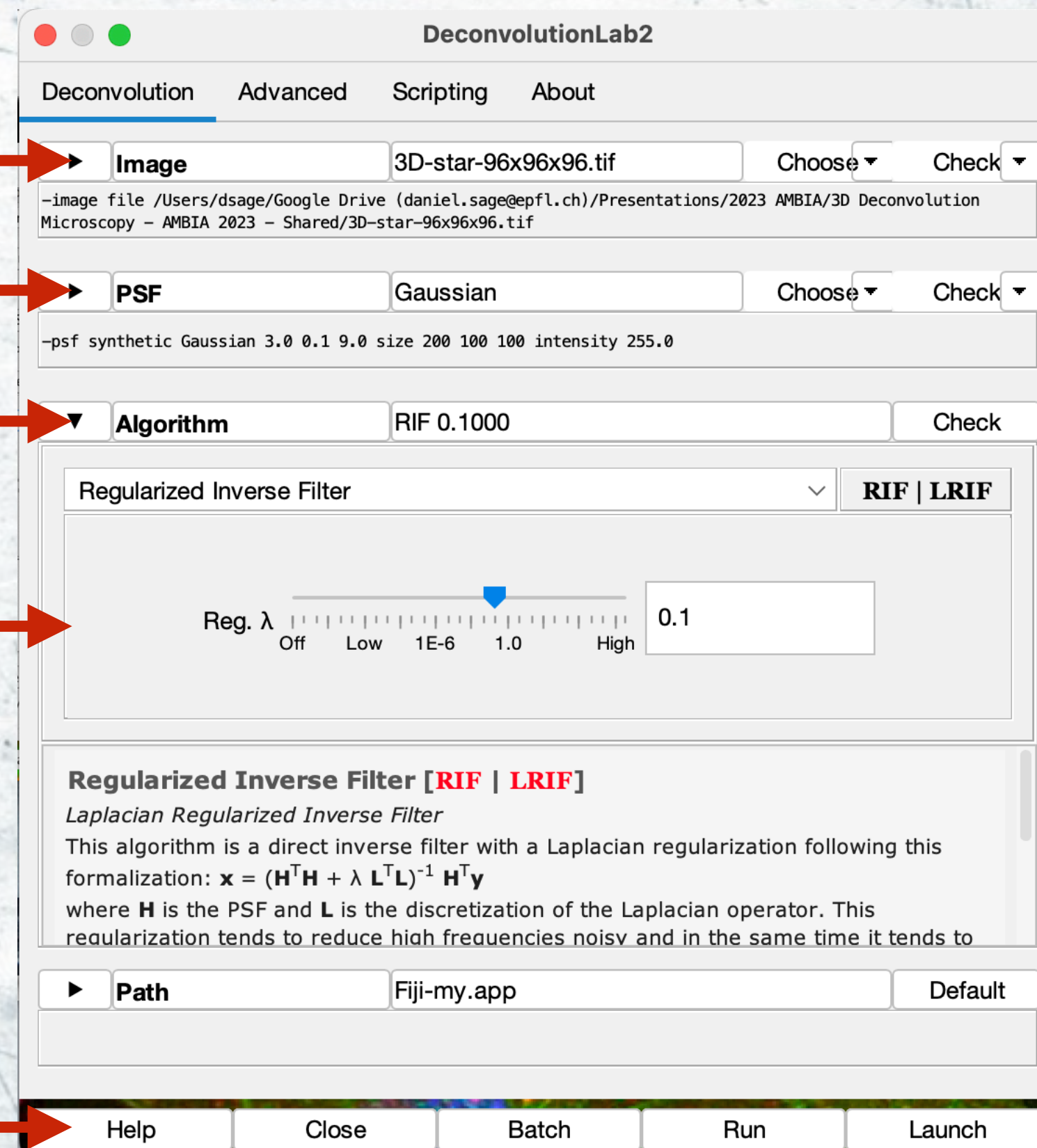
1) Select an image

2) Select a PSF

3) Choose an algo

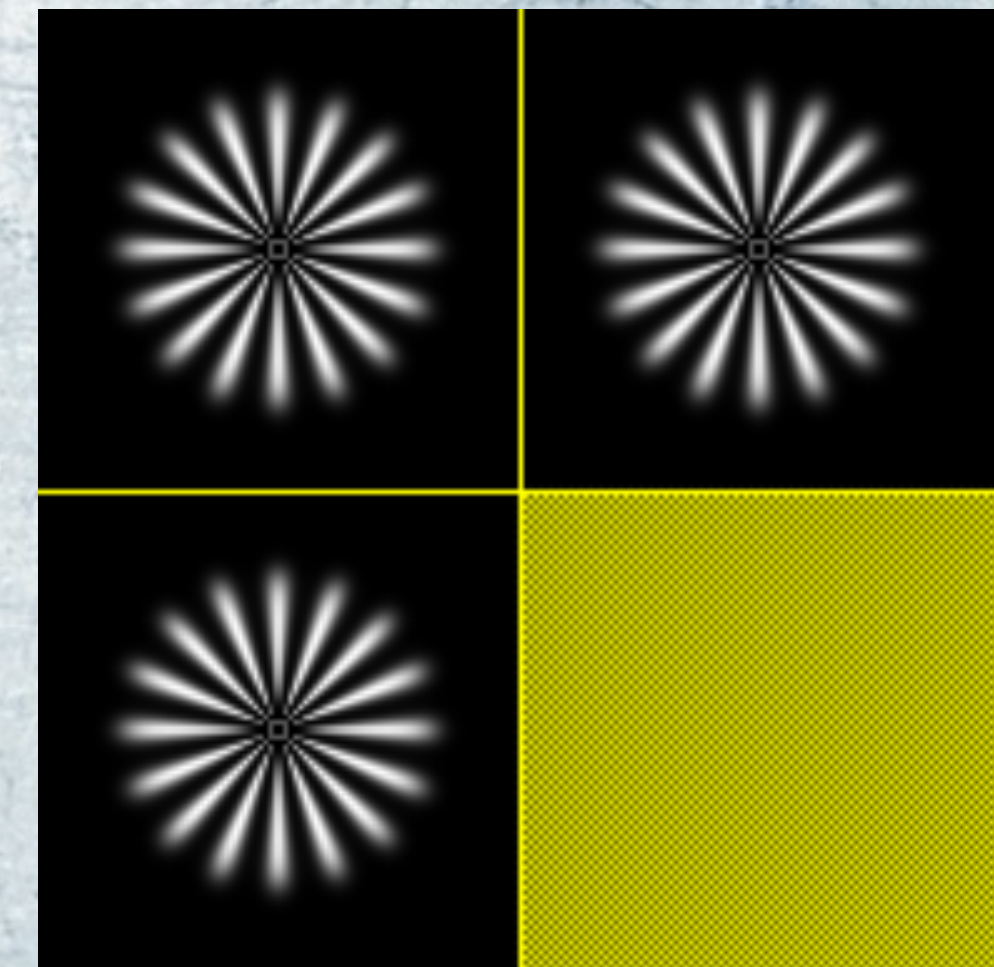
4) Set parameters

5) Run or launch



visualization

3D-star-96x96.tif



Output

Stack | MIP | Orthoview

<http://bigwww.epfl.ch/deconvolution/deconvolutionlab2/>

DeconvolutionLab2

Point-Spread Function

<http://go.epfl.ch/ambia2026/>

Experimental PSF

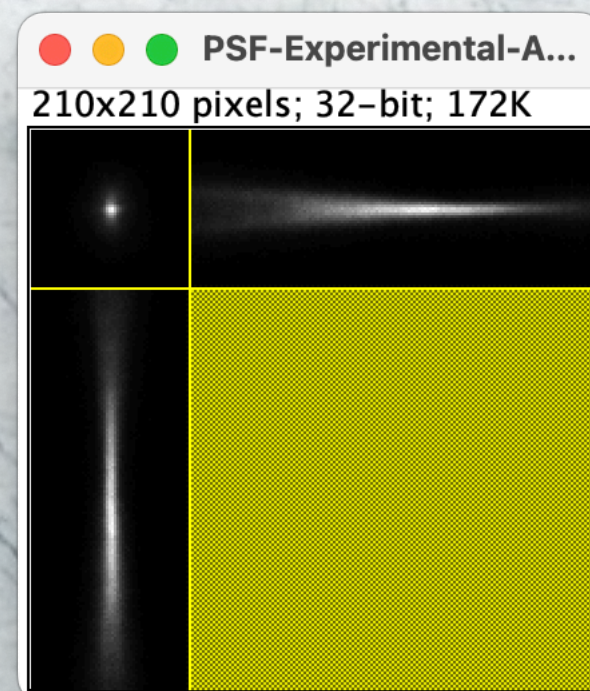
100nm bead shape

Select the PSF file on
DeconvolutionLab2

PSF-Experimental-2D.tif

or

PSF-Experimental-Astigmatism.tif



Synthetic PSF

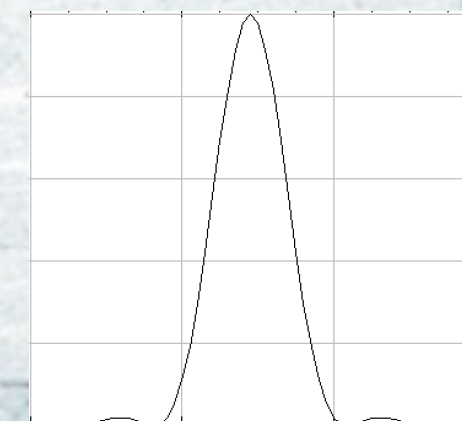
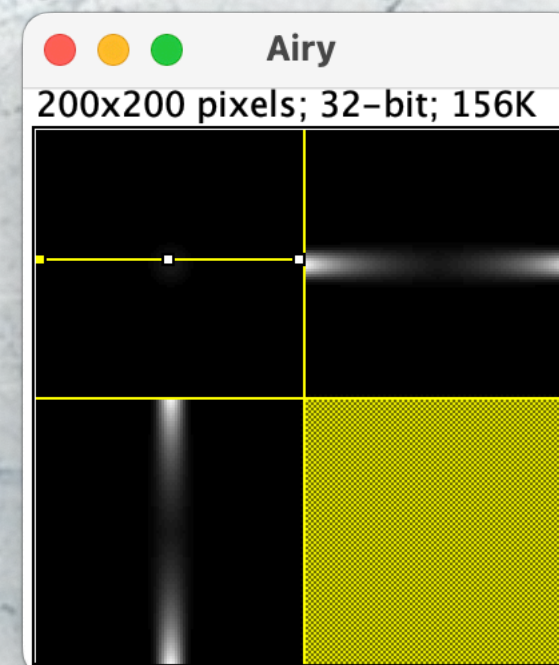
Simple shape

Choose "Get Synthetic" on
DeconvolutionLab2

Gaussian (default 100x100x100)

or

Airy (default 100x100x100)



Theoretical PSF

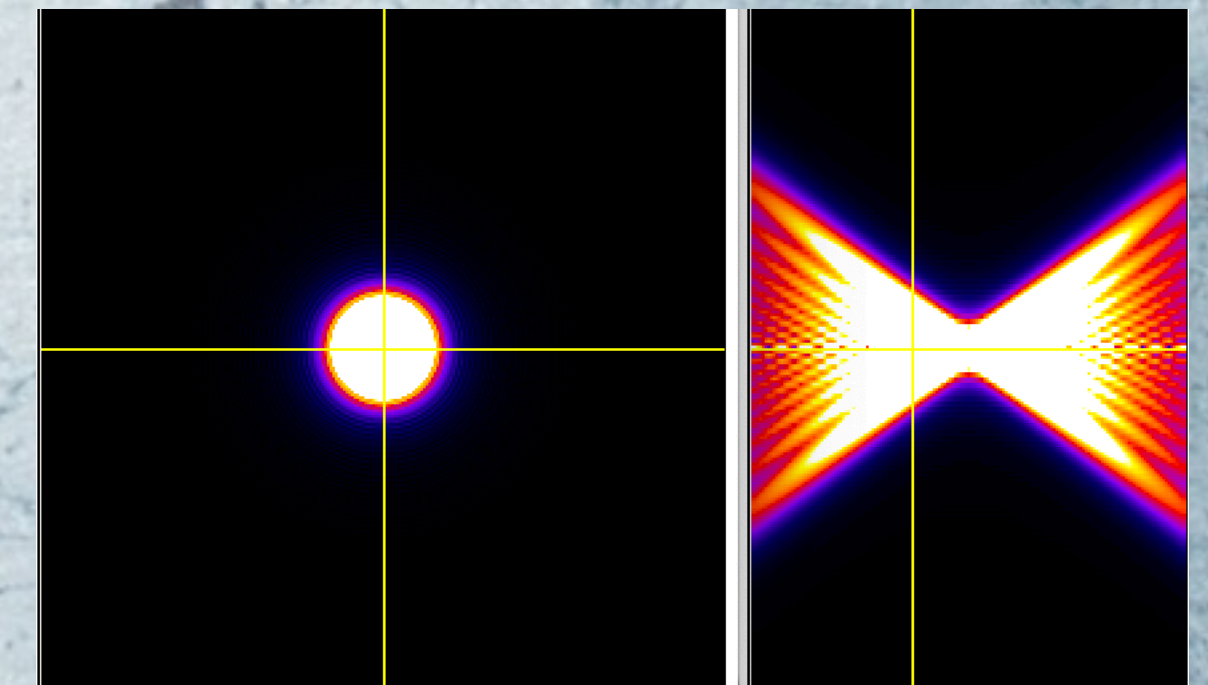
Microscopy shape

Use the plugins PSFGenerator.

Generator a PSF using

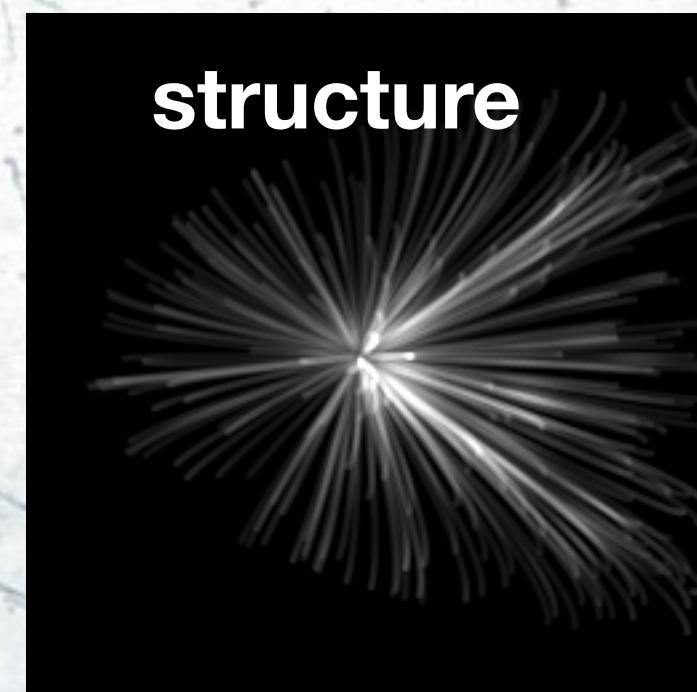
Born&Wolf (default values)

Save it as a TIF stack (32 bits)



DeconvolutionLab2

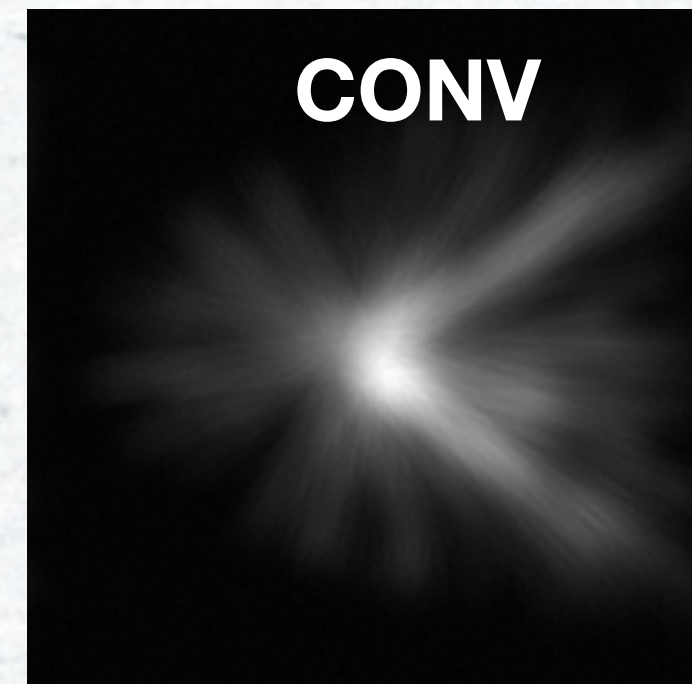
Inverse Filter



structure.tif
256x256



PSF
Airy-PSF.tif
128x128



CONV
256x256

Simulation

1. No noise, perfect PSF

Prepare an input image by convolving the structure with PSF using CONV.

Deconvolve it with DIV, NIF.

2. Small amount of noise, perfect PSF

Prepare an input image by convolving the structure with PSF using SIM with an additional Gaussian noise (mean=0, stdev=1).

Deconvolve it with DIV, NIF.

Deconvolve it with RIF, choose the appropriate lambda

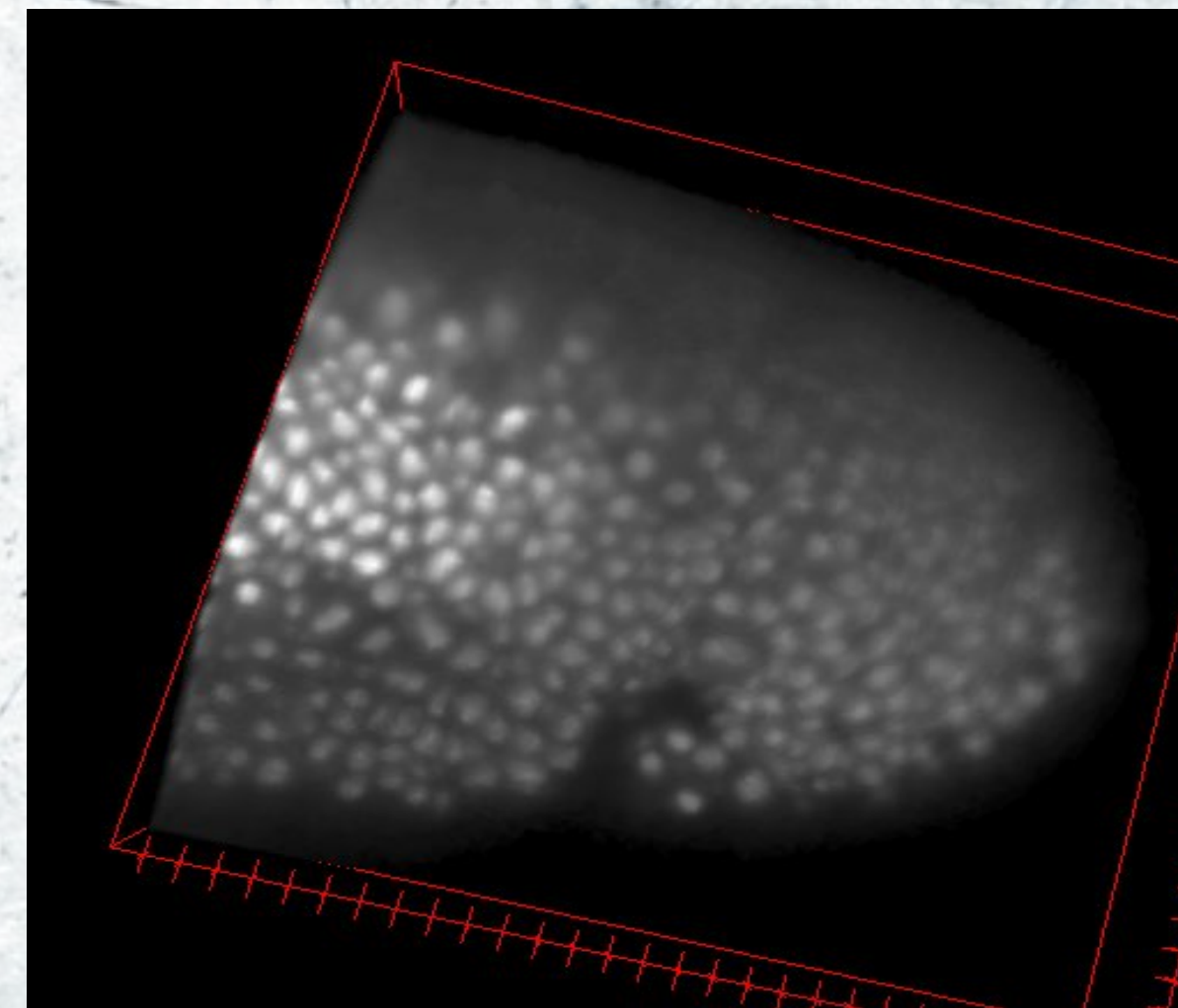
Deconvolution - Drosophila

A biologist wants to count the nuclei in the 3D image **drosophila**. The image present too much out-of-focus to be segmented with a simple method. Fortunately, she/he properly saved the optical parameters,

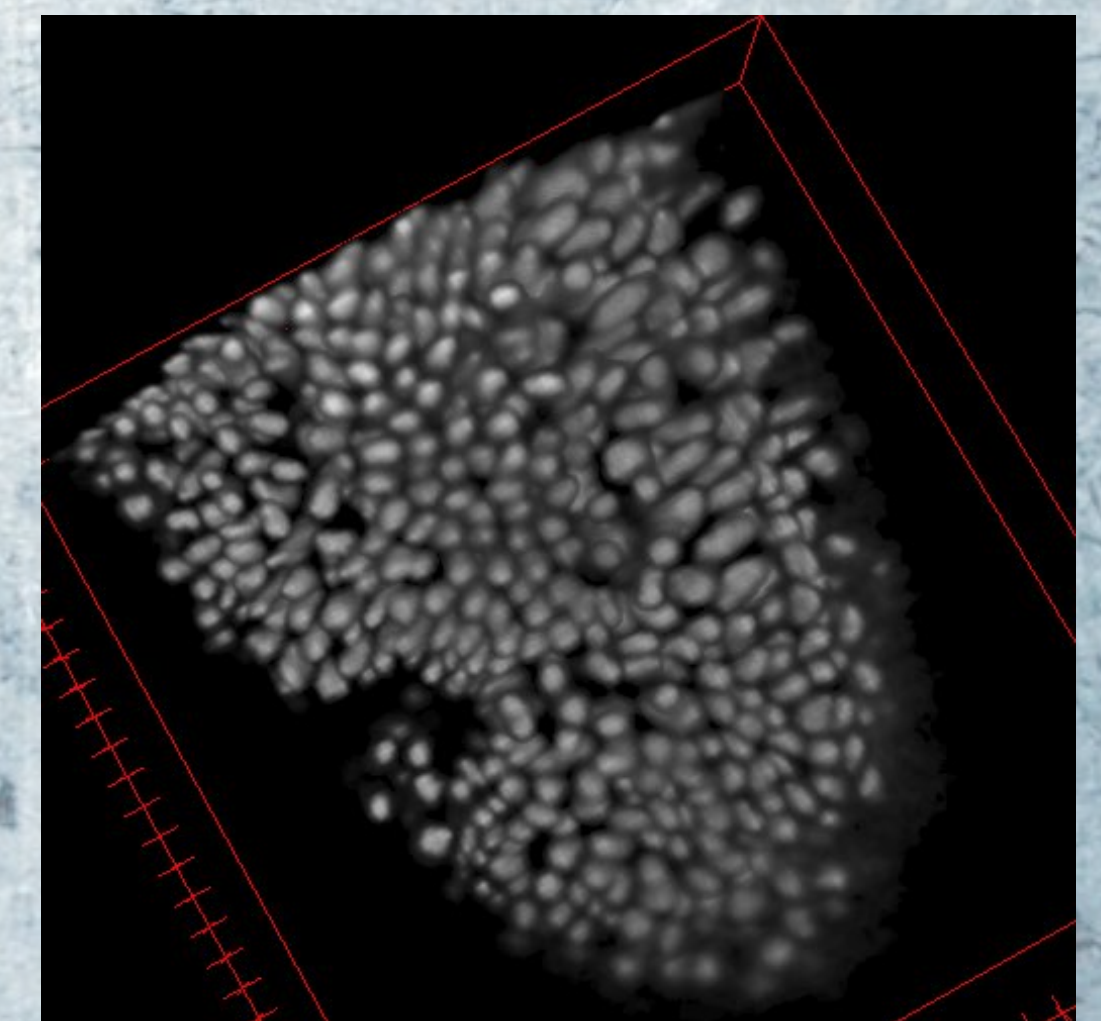
- wavelength=610 nm,
- NA=1.4
- refractive index immersion = 1.5,
- voxel size = 100 nm in all direction
- Model: Born&Wolf

Help him/her to generate a PSF and to deconvolve this image to remove the out-of-focus.

Input 300x275x50

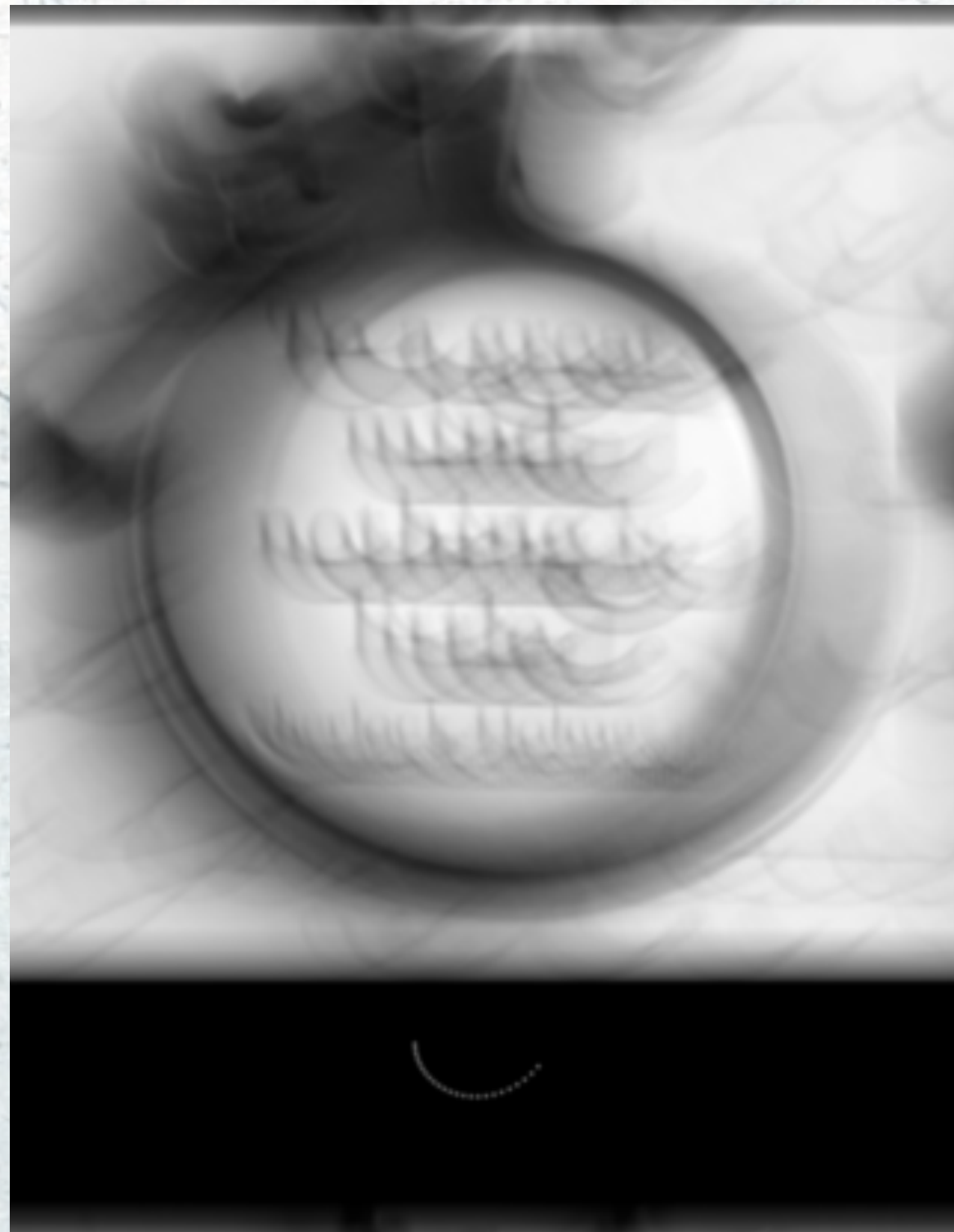


Desired output

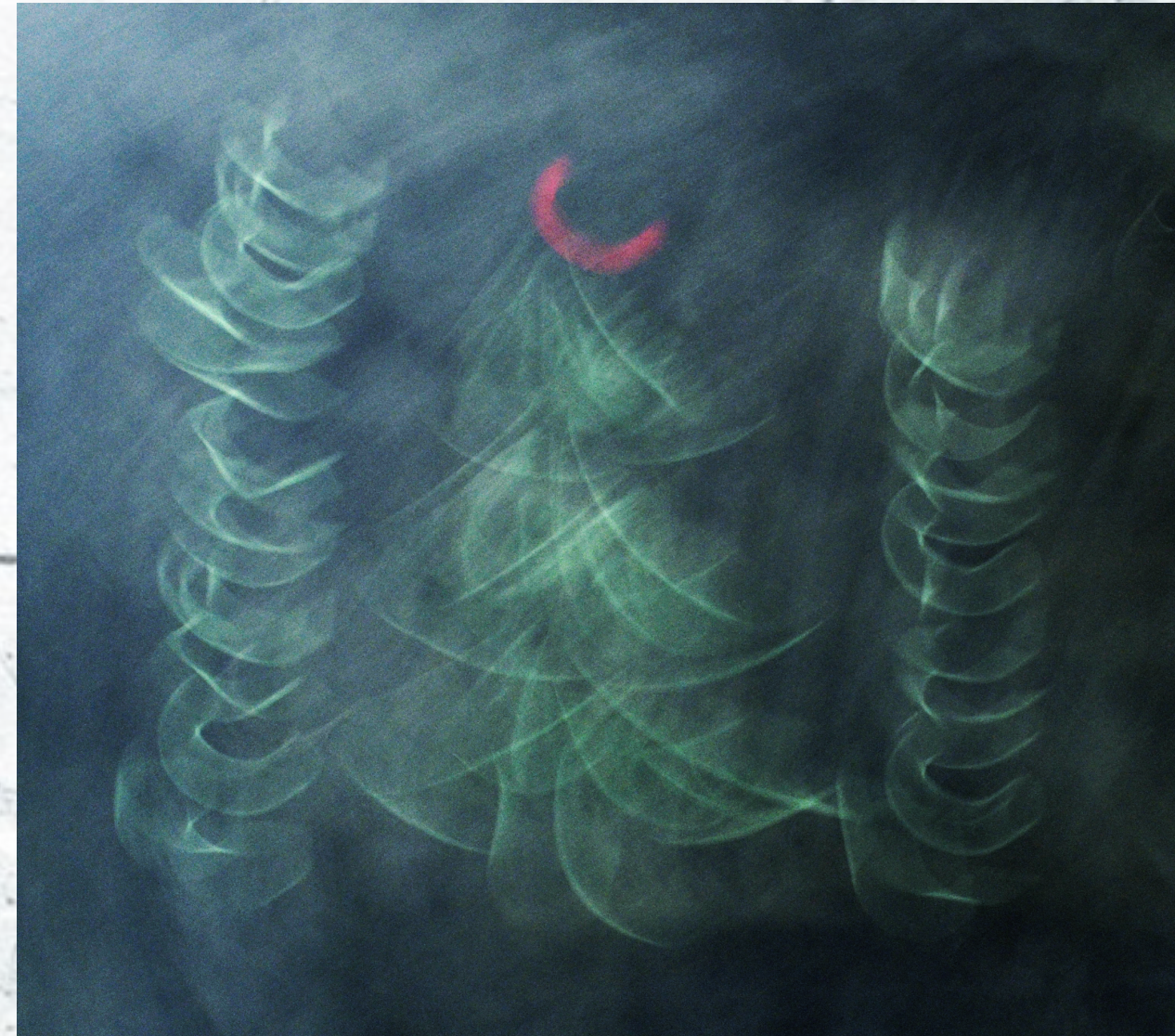


Save Sherlock Holmes!

quote.tif (simulation)



blackboard.tif (real)



moving-text.tif (real)



Challenging tasks

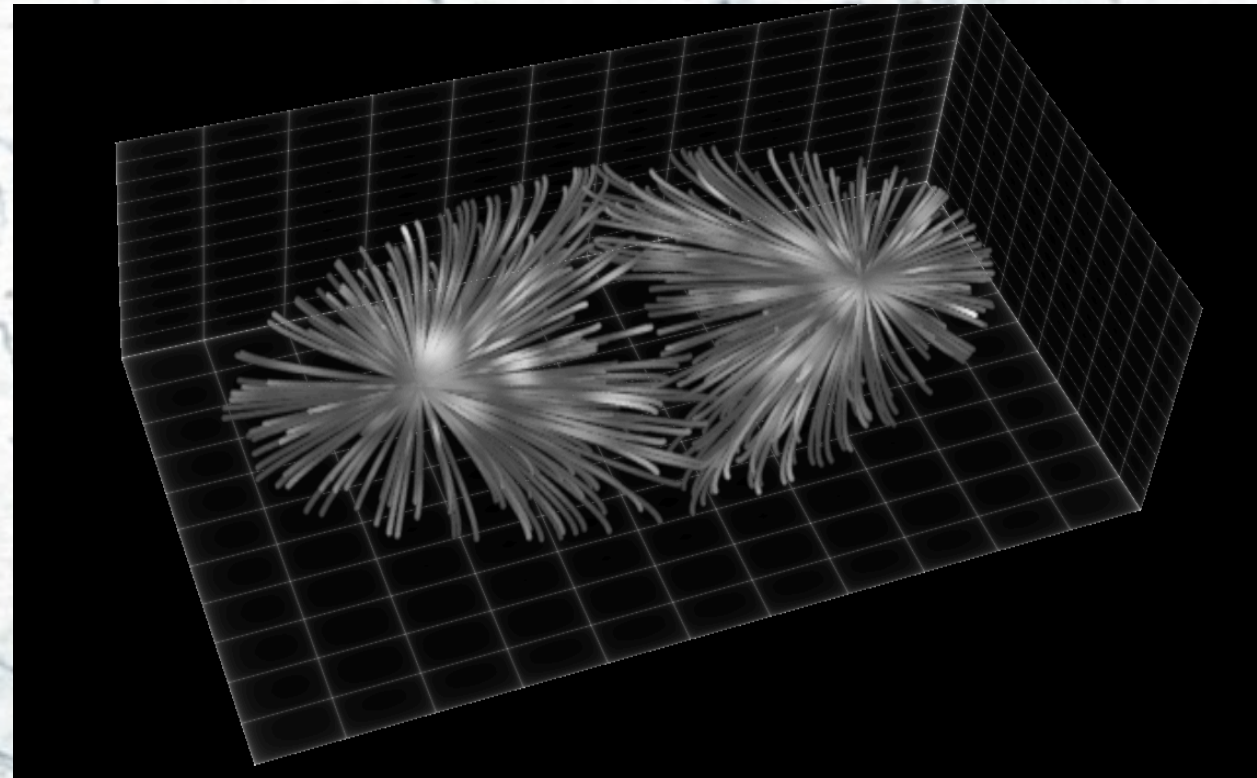
Could help Sherlock Holmes to read the text written in these images?

These three images were badly recorded by a moving camera:

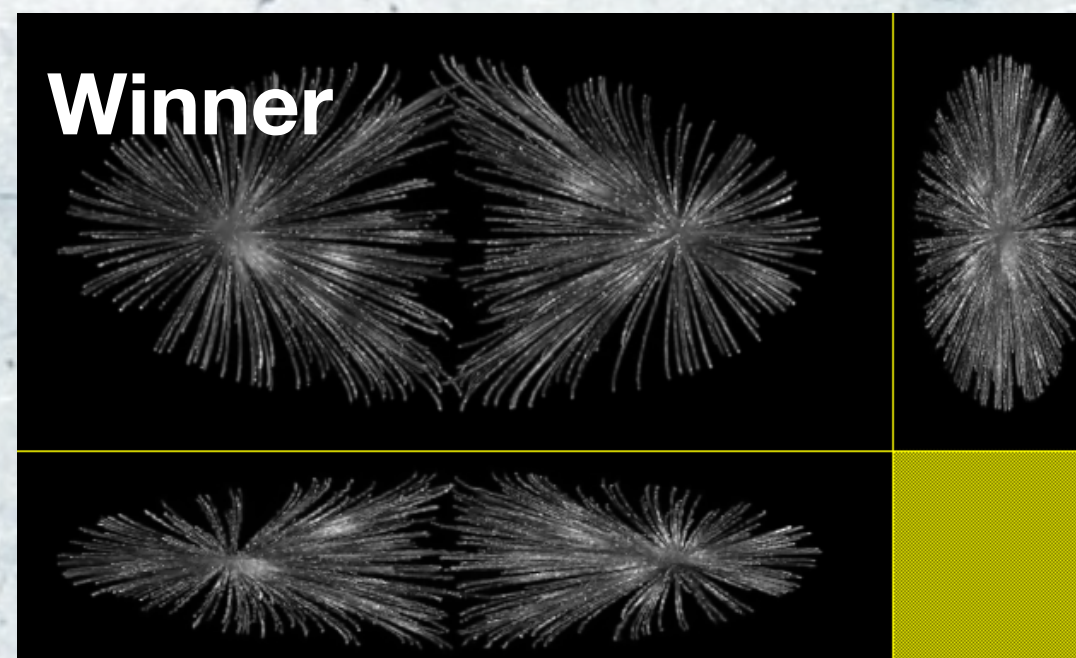
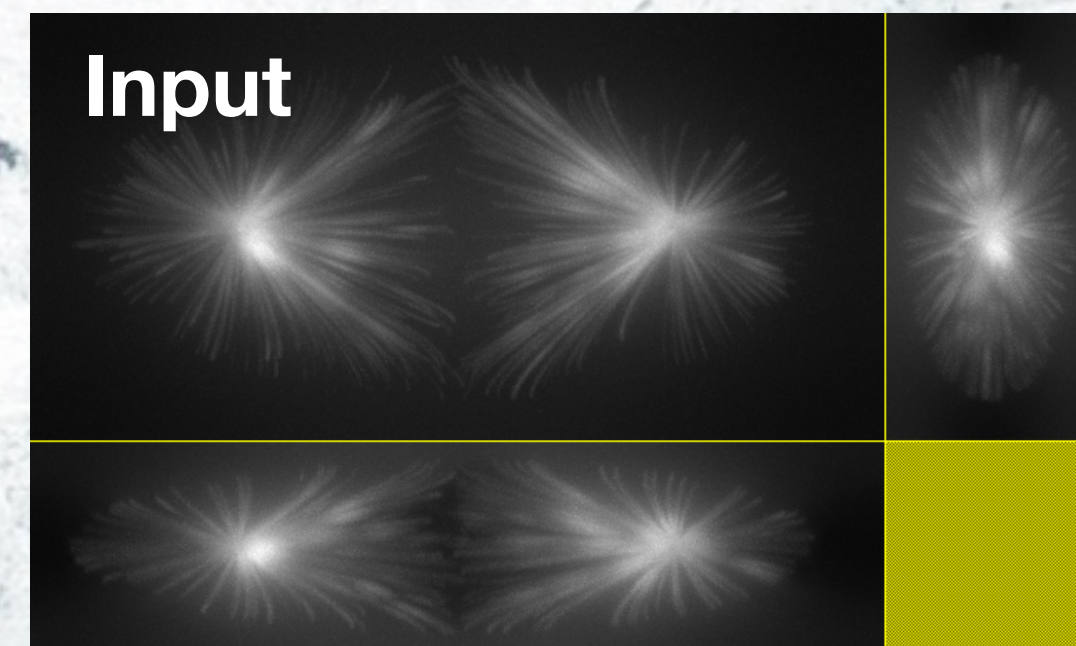
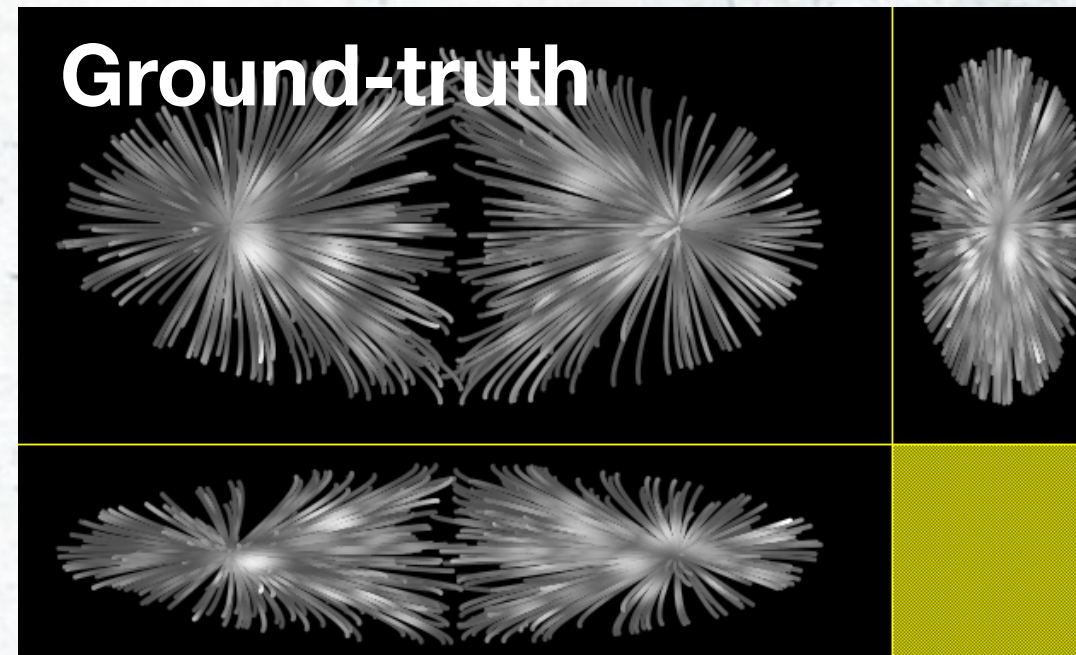
They have a strong motion blur but no additional information have been saved!

DeconvolutionLab2

Challenge 2014 GFP



In the ISBI challenge 2014, we asked participants to deconvolve an image **input** using a provided **PSF**. The dataset is a realistic 3D simulation of a network of microtubules during the mitosis. The PSF is a theoretical PSF. In order to assess deconvolution algorithms (SNR), we provide, here the 3D image **ground-truth**. We have the resulting image of the **winner** of the challenge and the one obtained by the software **huygens**.



Using DeconvolutionLab2, run different algorithms. Set the ground-truth image as Reference and select **show stats** in the Controller panel.

For each of the following images, compute the SNR with the reference and display a MIP image (maximum intensity projection)

- reference
- input
- winner
- huygens
- output of RIF 0.0005
- output of RL10

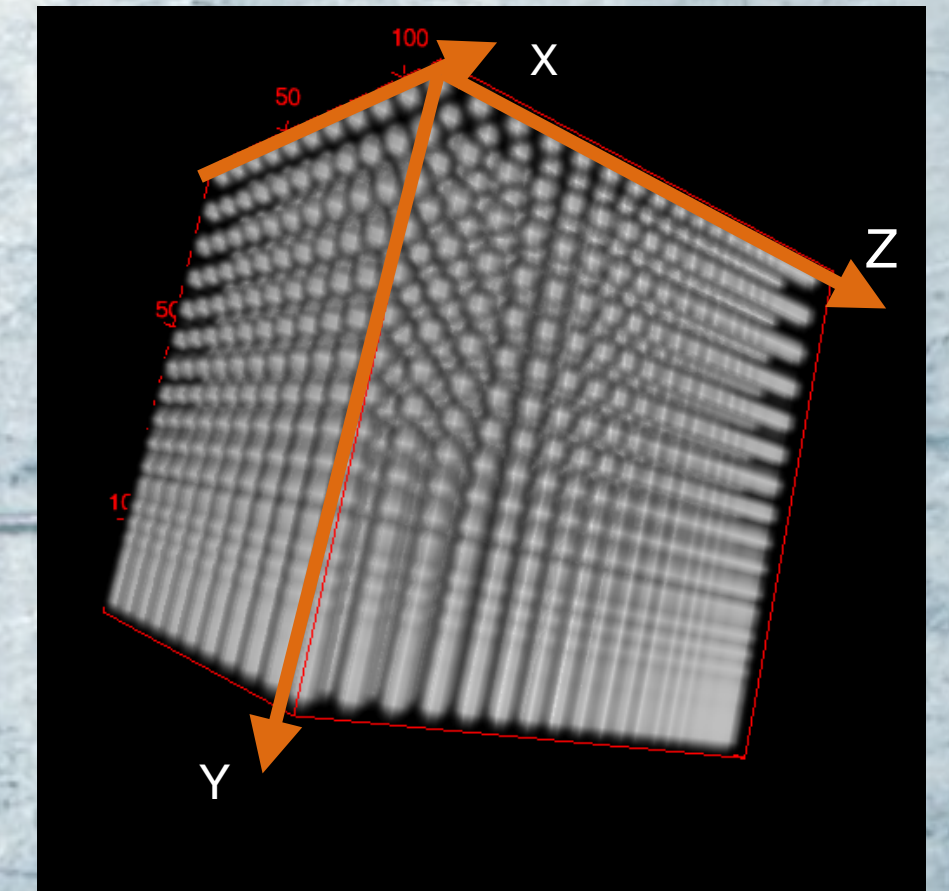
Try different algorithms and give your best result.

DeconvolutionLab2

Scripting

Here, we want to illustrate the effect of the FWHM of the PSF on the gain of resolution. This experience is tedious to do through the GUI, we prefer to do it using by scripting DeconvolutionLab2 is ImageJ Macro, Java, or Matlab. Here we show an example in Macro language

1. Generate a Cube of Spherical beads in a volume 128x128x128
2. Generate a non-isotropic Gaussian PSF ($\sigma_x = \sigma_y = 1.2$, $\sigma_z = 128$)
3. Create a simulation input (SIM) by convolving the input image with the PSF and adding Gaussian noise (mean = 0, sigma = 1.5)



```
dir = "/Users/dsage/Desktop/"
image = " -image synthetic CubeSphericalBeads 3.0 0.1 8.0 16.0 size 128 128 100 intensity 255.0"
psf = " -psf synthetic Gaussian 1.2 1.2 3.6 size 128 128 128 intensity 255.0"
parameters = " -out stack REF -path " + dir
run("DeconvolutionLab2 Run", image + psf + " -algorithm I " + parameters)

image = " -image file " + dir + "REF.tif"
parameters = " -out stack SIM -path " + dir
run("DeconvolutionLab2 Run", image + psf + " -algorithm SIM 0 1.5 0" + parameters)
```

Try to continue this script to show the effect of the PSF on the resolution.
(Refer to course slide)

DeconvolutionLab2